POLI 784 - Problem Set #3

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Question 1

Part A

Given the following regression:

$$y_i = \beta_0 + \beta_1 x_i + e_i$$

β_0 represents the intercept of the regression line, which is the expected value of y_i when x_i is equal to zero. β_1 represents the slope of the regression line, which indicates how y_i changes for a one-unit change in x_i. e_i represents the error term, which is the variation in y_i that cannot be explained by the linear relationship with x_i. The reasons that β_1 is not indexed by i is because it represents the consistent effect across all observations, meaning that the effect of x_i on y_i is assumed to be the same for all individuals in the sample. This indicates the structural restrictions of an OLS regression estimate, which assumes that the relationship between the independent variable and the dependent variable is linear and constant across all observations.

Part B

i) When $x' = x - 10$, and $x = x' + 10$, the new regression equation becomes:

$$y_i = \beta_0 + \beta_1(x' + 10) + e_i = \beta_0 + 10\beta_1 + \beta_1 x_i + e_i$$

β_0' becomes $\beta_0 + 10\beta_1$ and β_1' remains the same as β_1. This is because the intercept is changed by the amount of the shift in x, while the slope remains unchanged since it still represents the same change in y for a one-unit change in x.

ii) When $x' = 10x$, and $x = \frac{x'}{10}$, the new regression equation becomes:

$$y_i = \beta_0 + \beta_1 \left(\frac{x'}{10} \right) + e_i = \beta_0 + \frac{\beta_1}{10} x' + e_i$$

β_0' stays as β_0 and β_1' becomes $\frac{\beta_1}{10}$. This is because the slope is changed by the factor of the scaling of x, while the intercept remains unchanged since it is not affected by scaling.

iii) When $x' = 10x - 10$, and $x = \frac{x'}{10} + 1$, the new regression equation becomes:

$$y_i = \beta_0 + \beta_1 \left(\frac{x'}{10} + 1 \right) + e_i = \beta_0 + \beta_1 + \frac{\beta_1}{10} x' + e_i$$

β_0' becomes $\beta_0 + \beta_1$ and β_1' becomes $\frac{\beta_1}{10}$. This is because the intercept is changed by the amount of the shift in x, while the slope is also changed by the factor of the scaling of x.

Part C

i) When $y'' = y + 10$, and $y = y'' - 10$, the new regression equation becomes:

$$y_i'' - 10 = \beta_0 + \beta_1 x_i + e_i \implies y_i'' = \beta_0 + 10 + \beta_1 x_i + e_i$$

β_0'' becomes $\beta_0 + 10$ and β_1'' remains the same as β_1. This is because the intercept is changed by the amount of the shift in y, while the slope remains unchanged since it still represents the same change in y for a one-unit change in x.

ii) When $y'' = 5y$, and $y = \frac{y''}{5}$, the new regression equation becomes:

$$\frac{y_i''}{5} = \beta_0 + \beta_1 x_i + e_i \implies y_i'' = 5\beta_0 + 5\beta_1 x_i + 5e_i$$

β_0'' becomes $5\beta_0$ and β_1'' becomes $5\beta_1$. This is because both the intercept and the slope are changed by the factor of the scaling of y, since they both represent the expected value of y for a given value of x.

iii) When $y'' = 5y + 10$, and $y = \frac{y'' - 10}{5}$, the new regression equation becomes:

$$\frac{y'' - 10}{5} = \beta_0 + \beta_1 x_i + e_i \implies y_i'' = 5\beta_0 + 10 + 5\beta_1 x_i + 5e_i$$

β_0'' becomes $5\beta_0 + 10$ and β_1'' becomes $5\beta_1$. This is because the intercept is changed by both the shift and the scaling of y, while the slope is only changed by the factor of the scaling of y.

Part D

When x_i is shifted by a constant c, the intercept β_0 will change but the slope β_1 will remain the same. When x_i is scaled by a constant c, the slope β_1 will change by a factor of $\frac{1}{c}$, while the intercept β_0 will remain unchanged. When y_i is shifted by a constant c, the intercept β_0 will change by the same amount, while the slope β_1 will remain unchanged. When y_i is scaled by a constant c, both the intercept β_0 and the slope β_1 will change by a factor of c. However, all these operations don't affect the substantive conclusions about the relationship between x_i and y_i, as they only change the scale or the location of the regression line, but not the underlying linear relationship between the variables.

Question 2

Part A

i.

This means that ϵ_i represents the true error term, or the structural error, which captures all unobserved factors that affect the dependent variable (Y) but are not included in the model. Conversely, $\hat{\epsilon}_i$ represents the residuals, which are the differences between the observed values of Y and the predicted values based on the estimated regression line. The residuals are calculated from the sample data and are used to assess the fit of the model, while the error term is a theoretical construct that we cannot directly observe but is crucial for understanding the underlying data-generating process.

ii. $E[\hat{\epsilon}_i|D_i] = 0$ holds as a mathematical property of the fitted regression. In the predictive framework discussed in the intro to regression class slides, the statistical error is defined as the deviation of the outcome from its conditional expectation, $\epsilon = Y - E[Y|X]$, which implies that $E[\epsilon|X] = 0$ by definition of the conditional expectation. Because OLS estimates the best linear predictor of Y given the regressors by minimizing the sum of squared residuals, the fitted residuals inherit this property. The first-order conditions imply that the residuals have mean zero and are orthogonal to the included regressor D_i. Therefore $E[\hat{\epsilon}_i|D_i] = 0$ holds as a mathematical consequence of how the regression is defined, rather than as an assumption about the structural data-generating process.

iii. $E[\epsilon_i|D_i] = 0$ holds in the structural model only under the assumption of exogeneity. In the structural or causal framework discussed in the slides, the error term represents unobserved determinants of the outcome, and the condition $E[\epsilon_i|D_i] = 0$ requires that the treatment variable D_i is uncorrelated with these unobserved factors. This assumption would hold, for example, if treatment were assigned as if randomly, so that learning the value of D_i provides no information about the structural error. When the exogeneity condition holds, the OLS estimator recovers the true causal parameter τ in the structural model, so the estimated coefficient on D_i can be interpreted as the causal effect of the treatment on the outcome. If the condition fails, such as when omitted variables are correlated with D_i, the OLS estimate of τ will generally be biased.

Part B

We are given that the correct model is:

$$Y_i = \mu + \tau D_i + \beta U_i + \epsilon_i$$

The sample mean would then be:

$$\bar{Y} = \mu + \tau \bar{D} + \beta \bar{U} + \bar{\epsilon}$$

Subtracting the sample mean from the model, we get:

$$Y_i - \bar{Y} = (\mu + \tau D_i + \beta U_i + \epsilon_i) - (\mu + \tau \bar{D} + \beta \bar{U} + \bar{\epsilon})$$

Distributing the subtraction, we have:

$$Y_i - \bar{Y} = \mu + \tau D_i + \beta U_i + \epsilon_i - \mu - \tau \bar{D} - \beta \bar{U} - \bar{\epsilon}$$

The μ terms cancel out, leaving us with:

$$Y_i - \bar{Y} = \tau D_i + \beta U_i + \epsilon_i - \tau \bar{D} - \beta \bar{U} - \bar{\epsilon}$$

Rearranging the terms, we get:

$$Y_i - \bar{Y} = \tau D_i - \tau \bar{D} + \beta U_i - \beta \bar{U} + \epsilon_i - \bar{\epsilon}$$

Next, we can factor out τ and β from the respective terms:

$$Y_i - \bar{Y} = \tau(D_i - \bar{D}) + \beta(U_i - \bar{U}) + (\epsilon_i - \bar{\epsilon})$$

Part C

From the bivariate OLS estimator, we have:

$$\hat{\tau} = \frac{\sum_{i=1}^{N}(Y_i - \bar{Y})(D_i - \bar{D})}{\sum_{i=1}^{N}(D_i - \bar{D})^2}$$

From Part B, we showed that:

$$Y_i - \bar{Y} = \tau(D_i - \bar{D}) + \beta(U_i - \bar{U}) + (\epsilon_i - \bar{\epsilon})$$

Substituting this expression into the OLS estimator, we get:

$$\hat{\tau} = \frac{\sum_{i=1}^{N}\left[\tau(D_i - \bar{D}) + \beta(U_i - \bar{U}) + (\epsilon_i - \bar{\epsilon})\right](D_i - \bar{D})}{\sum_{i=1}^{N}(D_i - \bar{D})^2}$$

Distributing $(D_i - \bar{D})$ inside the brackets, we have:

$$\hat{\tau} = \frac{\tau\sum_{i=1}^{N}(D_i - \bar{D})^2 + \beta\sum_{i=1}^{N}(U_i - \bar{U})(D_i - \bar{D}) + \sum_{i=1}^{N}(\epsilon_i - \bar{\epsilon})(D_i - \bar{D})}{\sum_{i=1}^{N}(D_i - \bar{D})^2}$$

Separating the fraction into three terms, we get:

$$\hat{\tau} = \tau\frac{\sum_{i=1}^{N}(D_i - \bar{D})^2}{\sum_{i=1}^{N}(D_i - \bar{D})^2} + \beta\frac{\sum_{i=1}^{N}(U_i - \bar{U})(D_i - \bar{D})}{\sum_{i=1}^{N}(D_i - \bar{D})^2} + \frac{\sum_{i=1}^{N}(\epsilon_i - \bar{\epsilon})(D_i - \bar{D})}{\sum_{i=1}^{N}(D_i - \bar{D})^2}$$

Simplifying the first term, we obtain:

$$\hat{\tau} = \tau + \beta\frac{\sum_{i=1}^{N}(U_i - \bar{U})(D_i - \bar{D})}{\sum_{i=1}^{N}(D_i - \bar{D})^2} + \frac{\sum_{i=1}^{N}(\epsilon_i - \bar{\epsilon})(D_i - \bar{D})}{\sum_{i=1}^{N}(D_i - \bar{D})^2}$$

Part D

From Part C, we have that:

$$\hat{\tau} = \tau + \beta\frac{\sum_{i=1}^{N}(U_i - \bar{U})(D_i - \bar{D})}{\sum_{i=1}^{N}(D_i - \bar{D})^2} + \frac{\sum_{i=1}^{N}(\epsilon_i - \bar{\epsilon})(D_i - \bar{D})}{\sum_{i=1}^{N}(D_i - \bar{D})^2}$$

As the sample size becomes large, the last term converges to zero under the assumption that the structural error is uncorrelated with D_i. The second term converges to:

$$\beta \frac{\text{Cov}(D_i, U_i)}{\text{Var}(D_i)}$$

Thus, the probability limit of the OLS estimator is:

$$\hat{\tau} \to \tau + \beta \frac{\text{Cov}(D_i, U_i)}{\text{Var}(D_i)}$$

For omitted variable bias to be 0, the second term must equal zero. Since $Var(D_i) > 0$ whenever D_i varies in the sample, omitted variable bias is zero if either:

$$\beta = 0$$

or

$$\text{Cov}(D_i, U_i) = 0.$$

This means that omitted variable bias disappears if the omitted variable U_i has no effect on the outcome Y_i, or if U_i is uncorrelated with the regressor D_i. If neither of these conditions holds, the OLS estimator will generally be biased because part of the effect of U_i on Y_i is incorrectly attributed to D_i.

Question 3

In this exercise, we analyze data from a study by Santoro and Broockman (2022) examining the effect of conversations between strangers of opposing political parties on measures of polarization. During their conversation, subjects discussed what their perfect day would look like. Subjects assigned to treatment ($D_i = 1$) were informed prior to the start of the conversation that their partner was an out-partisan, while those assigned to control ($D_i = 0$) were not informed of this. The primary outcome of interest Y_i was self-reported warmth towards outpartisans measured on a feeling thermometer.

```r
# Load required packages
library("tidyverse")
library("estimatr")
library("sandwich")
library("interflex")
library("car")
library("glue")
```

```r
# Set random seed for reproducibility
set.seed(42)
```

Part A

Here, we compute an ATE point estimate using the Neyman estimator and corresponding analytical Neyman variance estimate. The formulae for these quantities are:

$$\hat{\tau}_{DIM} = \bar{Y}_1 - \bar{Y}_0 = \frac{1}{n_1} \sum_{i:D_i=1} Y_i - \frac{1}{n_0} \sum_{i:D_i=0} Y_i \tag{1}$$

$$\hat{\mathbb{V}}\left[\hat{\tau}_{DIM}\right] = \frac{s_1^2}{n_1} + \frac{s_0^2}{n_0} \tag{2}$$

where n_1 and n_0 are the number of units assigned to treatment and control, and s_1^2 and s_0^2 are the sample variances of the outcome variable in the treatment and control groups.

We subsequently compare these estimates to those produced using OLS regression with no covariates and the HC2 variance estimator.

```r
# Load data
df <- read.csv("data-2.csv")

# Rename outcome and treatment columns
colnames(df)[1:2] <- c("Y","D")

# Calculate ATE point estimate using Neyman estimator
n1 = sum(df$D)
n0 = sum(1-df$D)
tau_DIM <- mean(df$Y[df$D==1])-mean(df$Y[df$D==0])
var_DIM <- var(df$Y[df$D==1])/n1 + var(df$Y[df$D==0])/n0

# Calculate ATE using OLS and HC2 variance estimator
model <- lm_robust(Y ~ D, data = df, se_type = "HC2")
tau_OLS <- model$coefficients["D"]
var_HC2 <- model$std.error["D"]^2

# Display to console
glue("Neyman ATE point estimate: {round(tau_DIM,6)}")
```

```
Neyman ATE point estimate: 0.213034
```

```r
glue("Neyman variance: {round(var_DIM,6)}")
```

```
Neyman variance: 0.008334
```

```r
glue("OLS ATE point estimate: {round(tau_OLS,6)}")
```

```
OLS ATE point estimate: 0.213034
```

```r
glue("HC2 variance: {round(var_HC2,6)}")
```

```
HC2 variance: 0.008334
```

As shown above, we obtained identical estimates of the ATE and corresponding variance when using they Neyman estimator and OLS/HC2 estimator. This is not surprising, as the two approaches are algebraically equivalent in the case of a single covariate representing a binary treatment status. A proof of this is provided by Samii and Aronow (2012), which we briefly summarize below.

Assume we are fitting the following model:

$$Y_i = \alpha + \beta D_i + e_i \tag{3}$$

When D_i is a binary variable, the OLS estimates of α and β reduce to:

$$\hat{\alpha} = \bar{Y}_0 \tag{4}$$

$$\hat{\beta} = \bar{Y}_1 - \bar{Y}_0 \tag{5}$$

It can also be shown that for a binary treatment, the HC2 variance estimator reduces to:

$$\hat{\mathbb{V}}_{HC2}\left[\hat{\beta}\right] = \frac{1}{n_1}\frac{1}{n_1 - 1}\sum_{i:D_i=1}\hat{e}_i^2 + \frac{1}{n_0}\frac{1}{n_0 - 1}\sum_{i:D_i=0}\hat{e}_i^2 \tag{6}$$

where $\hat{e}_i$ represents the regression residual. This residual can be rewritten as:

$$\hat{e}_i = Y_i - \hat{\alpha} - \hat{\beta}D_i = Y_i - \bar{Y}_0 - \left(\bar{Y}_1 - \bar{Y}_0\right)D_i \tag{7}$$

Thus, $\hat{e}_i = Y_i - \bar{Y}_1$ when $D_i = 1$ and $\hat{e}_i = Y_i - \bar{Y}_0$ when $D_i = 0$. By substituting these expressions for $\hat{e}_i$ into Equation 6, we can rewrite the HC2 variance as:

$$\hat{\mathbb{V}}_{HC2}\left[\hat{\beta}\right] = \frac{1}{n_1}\frac{1}{n_1 - 1}\sum_{i:D_i=1}\left(Y_i - \bar{Y}_1\right)^2 + \frac{1}{n_0}\frac{1}{n_0 - 1}\sum_{i:D_i=0}\left(Y_i - \bar{Y}_0\right)^2 = \frac{s_1^2}{n_1} + \frac{s_0^2}{n_0} \tag{8}$$

Which is equivalent to the formula for they Neyman variance estimate shown in Equation 2.

Part B

Often we are fitting a model of the following form:

$$y_i = \mu + \tau d_i + \mathbf{x}_i^\top \ + e_i$$

The Frisch-Waugh-Lovell (FWL) theorem states that we can estimate τ via the following procedure:

1. Regress y_i on $\mathbf{x}_i$ and save the residuals $\hat{e}_{Y_i}$.
2. Regress d_i on $\mathbf{x}_i$ and save the residuals $\hat{e}_{D_i}$.
3. Regress $\hat{e}_{Y_i}$ on $\hat{e}_{D_i}$. The coefficient of $\hat{e}_{D_i}$ will be $\hat{\tau}$.

Here, we show that the estimate of $\hat{\tau}$ produced via the FWL procedure is equivalent to the estimate produced via multiple linear regression (MLR).

```
## Clean columns corresponding to covariates

# Began convo: Drop this variable since value is the same for everyone.
df <- df[, !(names(df) == "began_convo")]

# Gender: Set female to zero, everyone else as one.
df$gender <- 1 - as.numeric(df$gender == "Female")

# Note: we assume that all other covariates
# are numeric and continous, and thus do not modify them.


## Estimate tau via MLR
MLR_model <- lm_robust(Y ~ D +
                         age +
                         gender +
```

```r
                            partyideology +
                            pre_therm_voter_outparty_rescaled,
                          data = df,
                          se_type="HC2")

tau_MLR <- MLR_model$coefficients["D"]

## Estimate tau via FWL procedure

# Step 1: Regress Y on X
step1_model <- lm(Y ~ age +
                    gender +
                    partyideology +
                    pre_therm_voter_outparty_rescaled,
                  data = df)

df$FWL_resid_Y <- step1_model$residuals

# Step 2: Regress D on X
step2_model <- lm(D ~ age +
                    gender +
                    partyideology +
                    pre_therm_voter_outparty_rescaled,
                  data = df)

df$FWL_resid_D <- step2_model$residuals

# Step 3: Regress residuals from step 1 on residuals from step 2
FWL_model <- lm_robust(FWL_resid_Y ~ FWL_resid_D, data = df,se_type="HC2")
tau_FWL <- FWL_model$coefficients["FWL_resid_D"]

# Display to console
glue("MLR ATE Estimate: {round(tau_MLR,6)}")
```

```
MLR ATE Estimate: 0.326836
```

```r
glue("FWL ATE Estimate: {round(tau_FWL,6)}")
```

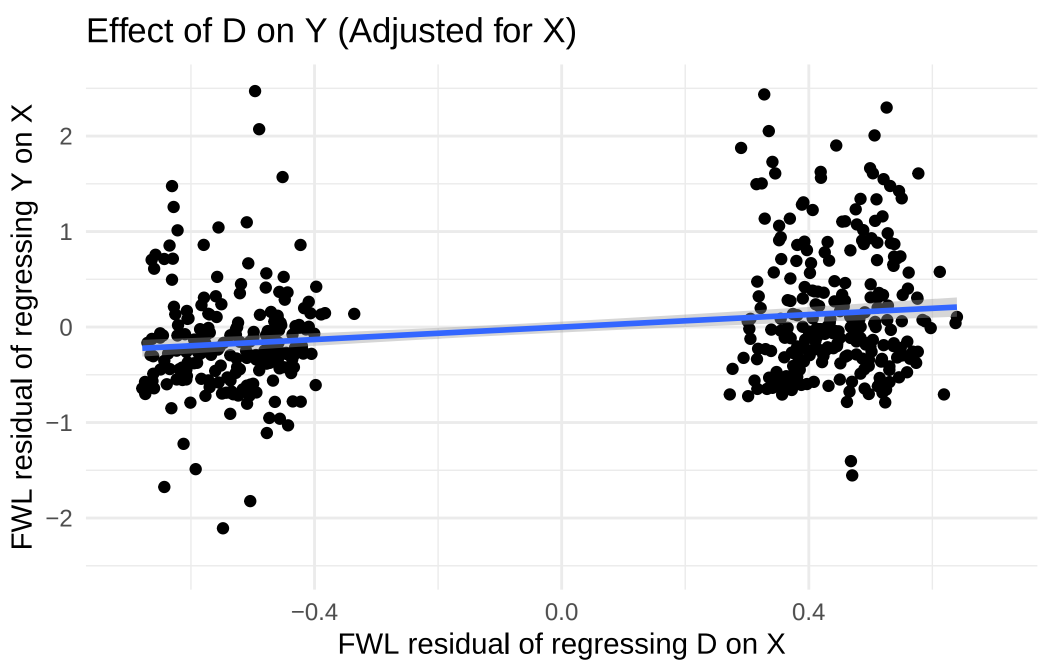
```
FWL ATE Estimate: 0.326836
```

Part C

Here, we use the residuals computed as part of the FWL procedure to examine the effect of D_i on Y_i before and after adjusting for covariates.

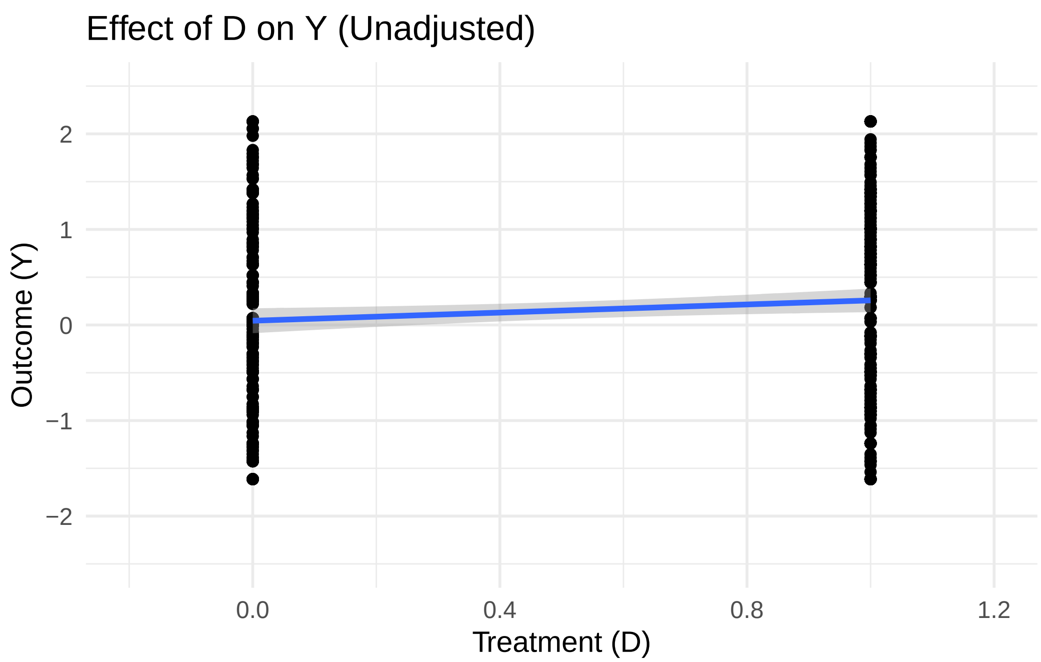
```
# Plot effect of D on Y after adjusting for X
p1 <- ggplot(df, aes(x = FWL_resid_D, y = FWL_resid_Y)) +
  geom_point() +
  geom_smooth(method = "lm_robust", method.args=list(se_type="HC2")) +
  xlim(-0.7,0.7) +
  ylim(-2.5,2.5) +
  labs(
    x = "FWL residual of regressing D on X",
    y = "FWL residual of regressing Y on X",
    title = "Effect of D on Y (Adjusted for X)"
  ) +
  theme_minimal()

# Display to console
p1
```

```
`geom_smooth()` using formula = 'y ~ x'
```



```r
# Plot effect of D on Y without adjusting for X
p2 <- ggplot(df, aes(x = D, y = Y)) +
  geom_point() +
  geom_smooth(method = "lm_robust", method.args=list(se_type="HC2")) +
  xlim(-0.2,1.2) +
  ylim(-2.5,2.5) +
  labs(
    x = "Treatment (D)",
    y = "Outcome (Y)",
    title = "Effect of D on Y (Unadjusted)"
  ) +
  theme_minimal()

# Display to console
p2
```

```
`geom_smooth()` using formula = 'y ~ x'
```



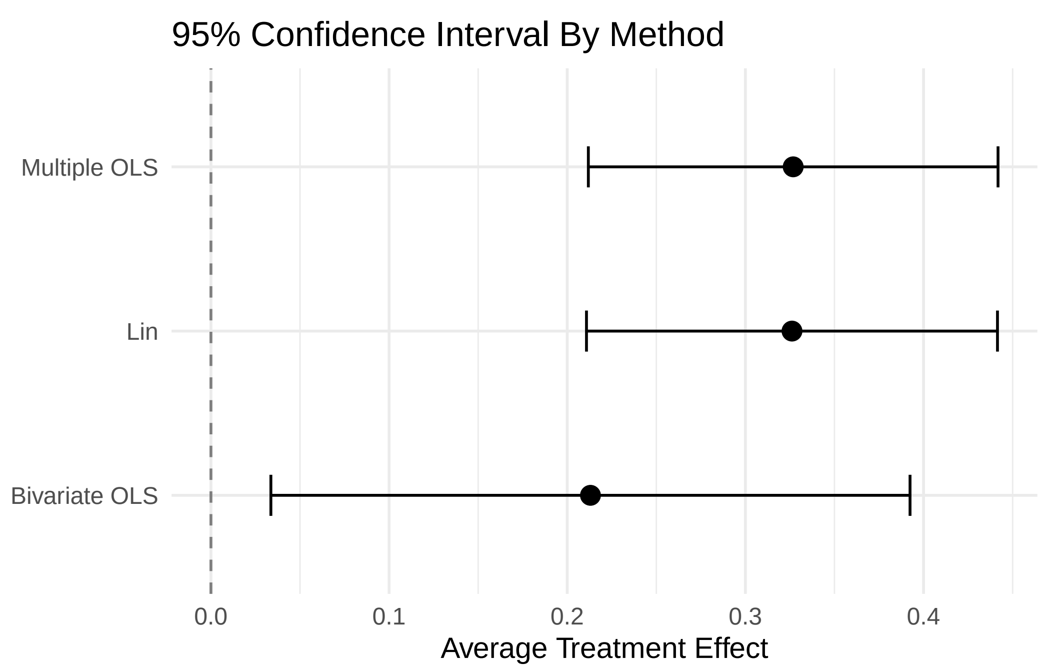
One notable difference in these two scatterplots is that the standard error on the regression line appears to be smaller for the plot that adjusts for covariates via the FWL procedure. This suggests that even in randomized experiments, it can often be helpful to adjust for baseline covariates in order to more efficiently estimate effects.

Part D

Here, we fit a Lin's regression model that captures interactions between the treatment and covariates. The estimated treatment effect and associated 95% confidence interval is subsequently compared against those obtained via multivariate OLS and bivariate OLS.

```
# Fit a Lin's regression model
lin_model <- lm_lin(Y ~ D,
                    covariates = ~age +
                      gender +
                      partyideology +
                      pre_therm_voter_outparty_rescaled,
                    data = df,
                    se_type = "HC2")

lin_estimate <- lin_model$coefficients["D"]
lin_LB <- lin_model$conf.low["D"]
lin_UB <- lin_model$conf.high["D"]

# Multiple OLS
multi_OLS_estimate <- MLR_model$coefficients["D"]
multi_OLS_LB <- MLR_model$conf.low["D"]
multi_OLS_UB <- MLR_model$conf.high["D"]

# Bivariate OLS
biv_OLS_estimate <- model$coefficients["D"]
biv_OLS_LB <- model$conf.low["D"]
biv_OLS_UB <- model$conf.high["D"]

# Dump results into a dataframe that we'll use for plotting
results_df <- tibble(
  method=c("Lin","Multiple OLS","Bivariate OLS"),
  estimate=c(lin_estimate,multi_OLS_estimate,biv_OLS_estimate),
  conf.low=c(lin_LB,multi_OLS_LB,biv_OLS_LB),
  conf.high=c(lin_UB,multi_OLS_UB,biv_OLS_UB)
)

# Create coefficient plot
```

```r
p <- ggplot(results_df, aes(x = estimate, y = method)) +
  geom_point(size = 3) +
  geom_errorbar(
    aes(xmin = conf.low, xmax = conf.high),
    width = 0.25
  ) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "gray50") +
  labs(
    x = "Average Treatment Effect",
    y = NULL,
    title = "95% Confidence Interval By Method"
  ) +
  theme_minimal()

# Display to console
p
```



Below, we compare the three regression approaches in terms of their efficiency, unbiasedness, and plausibility of underlying assumptions. Many of our interpretations are based on the papers by Freedman (2008) and Lin (2013). Please note that we are assuming a binary treatment throughout.

Efficiency

Lin's Regression: More efficient than bivariate OLS. Similar in efficiency to multivariate OLS in the above example, but in general should be as good or better than multivariate OLS because it can deal with heterogeneous treatment effects.

Multivariate OLS: More efficient than bivariate OLS and similar in efficiency to Lin's regression in the above example. However, it could be less efficient than one or both of these methods in other scenarios, such as in the presence of heterogeneous treatment effects or when there are nonlinear relationships between the outcome variable and pre-treatment covariates.

Bivariate OLS: Less efficient than Lin's regression. In the above example, it's also less efficient than multivariate OLS, but this may not hold in the presence of heterogeneous treatment effects.

Unbiasedness

All three estimators require SUTVA, positivity, and strong ignorability to hold in order to produce unbiased estimates of the ATE. Additional estimator-specific requirements are listed below.

Lin's Regression: Assumes a linear relationship between pre-treatment covariates and outcome, but is relatively robust to deviations from this assumption, especially when the sample size is large. Can accomodate heterogeneous treatment effects.

Multivariate OLS: Assumes a linear relationship between pre-treatment covariates and outcome. Also assumes a constant treatment effect.

Bivariate OLS: None. Equivalent to the Neyman difference-in-means estimator for binary treatment, so should be robust to heterogeneous treatment effects and nonlinear relationships between covariates and outcome.

Plausibility

In our specific case, the assumption of a constant treatment effect seems implausible. For example, we suspect that participants who lie in the middle of the political spectrum may respond differently to treatment than those at the extremes, who may be less tolerant of opposing viewpoints. As such, we would recommend using Lin's regression over multivariate OLS since the former can support heterogeneous treatment effects.

It also feels somewhat implausible to assume a linear relationship between the outcome variable (warmth towards outpartisans) and certain pre-treatment covariates such as age and party ideology. An easy way to address this issue would be to encode these variables as categorical predictors (e.g., "age 20-29", "age 30-39", etc.) rather than continuous predictors.

Part E

Here, we test the joint null hypothesis that the effect of D_i on Y_i equals zero for both women and men. In Part D, we fit a Lin's regression model that has the following form:

$$Y_i = \beta_0 + \beta_1 D_i + \cdots + \beta_7 \left(D_i \times \left(G_i - \bar{G}\right)\right) + \ldots$$

Where G_i is an binary variable that takes on a value of zero for women and one for men and $\bar{G}$ is the proportion of men in our sample. In this model, we can calculate conditional average treatment effects (CATE) among women and men as follows:

$$\text{CATE}_{\text{women}} = \beta_1 - \bar{G}\beta_7$$

$$\text{CATE}_{\text{men}} = \beta_1 + (1 - \bar{G})\beta_7$$

Our null hypothesis can therefore be formulated as:

$$H_0 : \beta_1 - \bar{G}\beta_7 = 0 \text{ and } \beta_1 + (1 - \bar{G})\beta_7 = 0$$

This hypothesis is evaluated using a Wald test implemented by the `linearHypothesis` function.

```r
# Calculate proportion of men in sample
G_bar <- mean(df$gender)

# Formulate null hypotheses
H0_women <- glue("D - {G_bar}*D:gender_c = 0")
H0_men <- glue("D + {1 - G_bar}*D:gender_c = 0")

linearHypothesis(lin_model,
                 hypothesis.matrix = c(H0_women,H0_men),
                 test = "Chisq")
```

```
Linear hypothesis test:
D - 0.49680170575693 D:gender_c = 0
D  + 0.50319829424307 D:gender_c = 0

Model 1: restricted model
```

```
Model 2: Y ~ D + age + gender + partyideology + pre_therm_voter_outparty_rescaled

  Res.Df Df  Chisq Pr(>Chisq)
1    461
2    459  2 31.728  1.289e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The results above indicate that we can be highly confident ($p < 1 \times 10^{-6}$) that the treatment has a nonzero effect among either women or men.

Part F

Here, we examine the moderating effect of baseline warmth towards outpartisans using (1) Lin's regression and (2) kernel regression.

Lin's regression

In Part D, we fit a Lin's regression model that has the following form:

$$Y_i = \beta_0 + \beta_1 D_i + \cdots + \beta_9 \left(D_i \times \left(W_i - \bar{W}\right)\right)$$

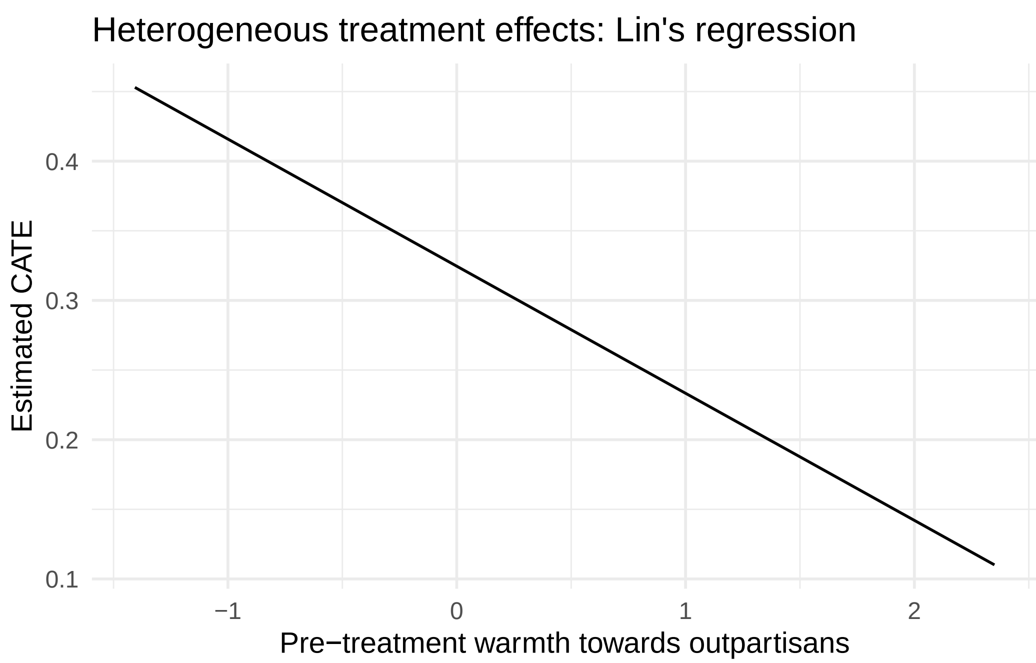
where W_i is the baseline (pre-treatment) measurement of warmth towards outpartisans and $\bar{W}$ is the average value of this covariate across the sample. The CATE given the value of this covariate is given by:

$$\text{CATE}(W = w) = \beta_1 + \left(w - \bar{W}\right) \beta_9$$

We can visualize this by plotting the CATE across the range of values observed in our data.

```r
# Get coefficient estimates
beta1 <- lin_model$coefficients["D"]
beta9 <- lin_model$coefficients["D:pre_therm_voter_outparty_rescaled_c"]
W_bar <- mean(df$pre_therm_voter_outparty_rescaled)

# Get range of values
W_min <- min(df$pre_therm_voter_outparty_rescaled)
W_max <- max(df$pre_therm_voter_outparty_rescaled)
W_vals <- seq(W_min,W_max,length.out=200)
```

```r
# Calculate CATE over range of values
CATE_W <- beta1 + beta9*(W_vals - W_bar)

# Create plot
p <- ggplot(data.frame(w=W_vals,CATE=CATE_W),aes(x = w, y = CATE)) +
  geom_line() +
  labs(
    x = "Pre-treatment warmth towards outpartisans",
    y = "Estimated CATE",
    title = "Heterogeneous treatment effects: Lin's regression"
  ) +
  theme_minimal()

# Display to console
p
```



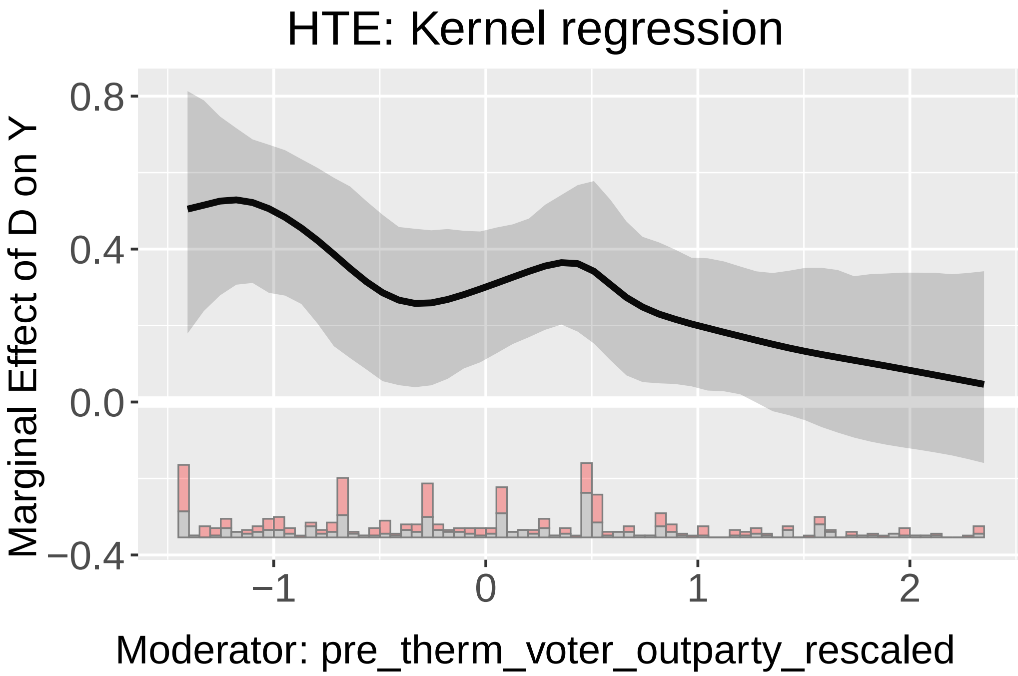
Kernel regression

We can also visualize the moderating effect of pre-treatment warmth towards outpartisans using nonparametric techniques such as kernel regression.

```r
## Fit kernel regression model with pre-treatment warmth as a moderator.

# Note that I'm manually setting the bandwidth because
# the default value appeared to be overfitting.
# Settled on a value of 0.3 based on the range of our moderator variable.

model_kernel <- interflex(
  Y = "Y",
  D = "D",
  X = "pre_therm_voter_outparty_rescaled",
  data = df,
  bw = 0.3,
  estimator = "kernel",
  vcov.type = "robust"
)
```

```
Baseline group not specified; choose treat = 0 as the baseline group.
Number of evaluation points:50
Parallel computing with 4 cores...
```

```r
# Plot results
plot(model_kernel,
     main = "HTE: Kernel regression")
```



Interpretation

The above results suggest that the treatment effect was reduced for those who had higher baseline measures of warmth towards outpartisans. One potential explanation for this is that those who had higher baseline warmth towards outpartisans might already have regular face-to-face interactions with people whose political beliefs differ from theirs, which would make the experiment "just another day" for them. In contrast, those who had low baseline warmth might live in communities without much political diversity, which would make the experiment a more abnormal and impactful experience.